

Open Selection Graph (OSG): a dataset for mapping observable populations and selection processes in science

Anton Waagaard

Dimensional Impact Lab

anton@dimactlab.org

Manuscript version 2.0, 23 August 2026

Abstract

Research selection leaves uneven public records. Submission lists may begin after editorial screening, funding records may cover only proposals that reached a panel, and patent data omit confidential applications. Open Selection Graph (OSG) organises these partial records around the populations and stages that can be observed. It links candidates and document versions to institutional rules, evaluations, decisions, lineage, and later outcomes across peer review, research funding, and patent examination. The dataset contains a 4,872-cycle observability census, audited OpenReview cohorts, public UKRI and SNSF panel records, a PANORAMA patent examination panel, and a privacy-minimised census of 4,518,254 published English-language US utility applications filed from 2004 through 2018.

Keywords: peer review; research evaluation; scientific selection; metascience; open data; observability; institutional design; research funding; patent examination; data provenance

1 Background & Summary

Peer-review datasets have made reviews, scores, decisions, and manuscript text available for computational and metascientific research. PeerRead assembled paper drafts, decisions, and review text from several computer-science venues [1]. NLPeer subsequently emphasised ethical sourcing, explicit licensing, multiple domains, manuscript versions, and structured representations [2]. Platform-native resources and newer derived corpora add scale and task diversity. Most organise the evidence around a paper, a review, or an accepted/rejected pair.

Many institutional questions concern the process that made these records observable. An OpenReview submission invitation, a Copernicus discussion paper, an eLife Reviewed Preprint, a post-publication F1000 article, an opt-in PLOS review history, a public grant award, and a published patent application enter the public record at different selection stages. Their architectures also differ. Competitive quotas, rolling thresholds, public discussion, publish-review-curate systems, fundable-band allocation, and patent prosecution generate distinct outcome states. A large review corpus may therefore leave its candidate population and selection stage unidentified.

Publication, funding, and patent examination share a relational structure. Candidates enter an institutional cycle governed by a policy. Evaluators produce evidence, decisions alter candidate states, and later records describe revision, allocation, legal prosecution, or downstream use. OSG represents this common structure and retains the native meanings of “accepted,” “fundable,” “awarded,” “rejected under §103,” and “allowed.” The three domains provide contrasting selection systems for studying observability, choice sets, staged evaluation, and outcome semantics.

OSG records institutional observability as part of the data model. Its starting question is: **what population is visible, from which stage, under which rules, and with which independently checkable denominator?** The answer determines whether a source supports entry-selection estimates, stage-conditional estimates, process description, portfolio description, or source auditing alone.

This design is also motivated by the fragmented public infrastructure around scholarly processes. Open-Review exposes Notes, Invitations, Groups, Edits, and reader permissions through versioned APIs [3–5]. Crossref exposes bibliographic metadata and relations; independent provider evidence establishes submission-population completeness [6]. OpenAlex supplies an open scholarly graph for downstream works and citations [7]. arXiv and Europe PMC expose versioned metadata and, where licensed, structured content [8,9]. Publisher and platform families expose different portions of editorial process through public pages, APIs, OAI-PMH, JATS XML, and Crossref relations [10–14]. Public funding portals and patent datasets expose still different choice sets and selection boundaries [15–18]. OSG joins these systems through a common graph and preserves their institutional differences as source-level data.

Five design elements organise the release.

1. It defines the gate cycle as the primary institutional unit.
2. It represents coverage, hidden stages, and denominator evidence as versioned tables and enforces them in analysis views.
3. It retains native stages, rubrics, policy states, and relations before normalisation.
4. It connects evaluations to document versions, downstream outcomes, research-funding rounds, and patent prosecution while preserving temporal order and domain-specific outcome meanings.
5. It provides a licence-aware, privacy-preserving release in which quarantine and `not_identified` are explicit data states.

2 Methods

2.1 Gate cycles and institutional architectures

A `gate_cycle` is the smallest venue, track, call, panel, journal-period, or policy-stable interval for which the applicable rules and observable population have coherent meanings. It belongs to a `gate` and links to a `policy_version`, one or more `coverage_observation` records, candidate-gate events, evaluations, decisions, and capacity observations. Candidate documents are subordinate objects with one or more versions.

The architecture registry distinguishes at least seven process families:

- **competitive quota**, in which candidates are compared for a limited set of places;
- **rolling threshold**, in which candidates are assessed against a standard without a fixed cohort quota;
- **access review followed by public discussion**, in which the public population starts only after a hidden access screen;
- **publish-review-curate**, in which publication and subsequent evaluation or curation are distinct stages;
- **post-publication review**, in which editorial screening precedes public review and versioning;
- **fundable-band or lottery allocation**, in which scientific eligibility and allocation may be distinct stages; and
- **patent prosecution or examination**, in which claims, office actions, amendments, and legal outcomes form a multi-turn process.

Architecture determines valid stage mappings, denominators, and comparative analyses. A generic `accepted` field may be derived for a declared estimand. Native state machines remain authoritative.

2.2 Observability grades

Each `source` × `gate_cycle` × `object_type` population receives one of five grades:

Grade U marks an unresolved population boundary. These rows support source audits, comparisons of institutional opacity, missingness research, and longitudinal monitoring of disclosure practices. Denominator-dependent rates exclude them.

Table 1. Observability grades and admissible uses.

Grade	Population boundary	Admissible use
A, entry-complete	The earliest submitted pool is independently verified, including observable withdrawals and negative outcomes	Entry-selection and stage-transition estimates

Table 1. Observability grades and admissible uses.

Grade	Population boundary	Admissible use
B, stage-complete	The public pool is verified after a named hidden screen	Selection estimates conditional on reaching that public stage
C, selected/opt-in history	Reviews or histories exist only for selected, published, or opted-in works	Descriptive evaluation and revision research
D, outcome registry	Winners, publications, or outcomes are visible without a comparable candidate pool	Portfolio and outcome description
U, unresolved	Denominator, stage, or source behaviour is unresolved	Source auditing and missingness research

Grades are local and time-varying. A provider may change platform conventions, public readers, APIs, or disclosure policy. An apparent A or B population is automatically downgraded when count coverage is below 0.95, when stage-flow identities fail, or when the earliest observable stage is ambiguous. OSG distinguishes two products that were previously conflated. The *observability census* contains 4,872 rows. This total includes 4,522 OpenReview policy and configuration surfaces. They are recorded as policy observations and excluded from process-cycle counts. The *process-level analytical atlas* contains 350 reconstructed cycles. Of these, 210 have verified Grade A/B denominators and 200 have a non-zero observable population. 61 cycles support a complete observed selection rate. A further 197 cycles support review-incidence measures against a verified public candidate denominator. Denominator validity, review coverage, and decision coverage are stored as separate properties.

Temporal-change coverage is limited to a repeated audit of this public OpenReview surface. Two snapshots, observed on 20 and 23 August 2026, produce 410 cycle-snapshot rows. Across that interval, 9 cycles changed their provider-reported public-note count and 0 changed observability grade. The resulting history covers short-interval change in the audited OpenReview cycles from 20 to 23 August 2026.

2.3 Native-first normalisation

Every normalised stage, outcome, criterion, scale, relation, field, and policy trait retains its native label or value. For example, `decision_event` stores both `outcome_native` and `outcome_normalized`; `evaluation` stores the native criterion, raw value, numeric representation where defensible, and native scale definition. A changed score scale creates a new rubric or policy version. Unsupported mappings remain native-only.

This rule prevents false equivalences such as treating “reviewed preprint,” “publish,” “major revision,” “fundable,” “withdrawn,” and “notice of allowance” as values on a universal acceptance scale. Broad constructs, including novelty/originality, significance/interest, soundness/evidence, clarity, reproducibility, ethics, confidence, and overall recommendation, are crosswalks that preserve the underlying native criteria.

2.4 Temporal truth

The schema distinguishes source creation, source modification, retrieval, observation, policy effectiveness, decision time, identity visibility, and outcome censoring. Point-in-time queries can therefore separate what was later known from what could have been known at a decision. A decision-time feature view excludes later manuscript versions, future citations, and identities revealed after review. Reference corpora for novelty measures are strictly prior to the target cohort.

2.5 Identification states

Each source has an explicit state: included, pointer-only, derived-only, quarantined, blocked by terms, blocked by coverage, or retired. Registered estimands return `identified`, `partially_identified`, or `not_identified` according to their required fields and admissible observability grades. This design keeps failed acquisitions and unsupported estimands visible in the census.

This distinction is important for public funding data. Allocation effects require assignment arms and an eligible choice set. Repeated awards identify award recurrence, whereas application recurrence requires applicant-level submission histories. Patent populations also require an explicit boundary because public sources omit confidential and non-published filings.

2.6 Partial identification and comparative support

OSG turns two common cautions into analysis tables. For every census cycle, **observability_partial_identification** records whether the entry population is observed, its identified value when available, and otherwise the no-assumption region implied by the visible stage. If the public population begins after an unobserved screen, the number of hidden candidates has no finite public-data upper bound and the entry-selection probability remains in $[0, 1]$. A companion table contains 34,104 sensitivity rows over seven declared public-stage capture fractions. Every row is labelled as a sensitivity scenario.

The **transportability_diagnostics** table evaluates all 120 source pairs using overlap in calendar time, observable population size, selection share, and review intensity, together with architecture compatibility and standardised differences. The table measures empirical support in these recorded dimensions. Domain coverage is incomplete, and joinable cycle-level policy and disclosure covariates are absent; the diagnostics record them as required-but-unmeasured effect modifiers. The current release supports conditional modelling with sensitivity analysis for some pairs and descriptive stratification for others. It finds insufficient support for direct source-pair comparison.

For content whose terms preclude redistribution, 684 release rows supply an HTTPS source, object locator, payload-derived SHA-256 verification hash, licence or terms field, and retrieval contract. The registry separates technical verifiability from redistribution rights.

2.7 Data sources and scope

2.7.1 Source census

Version 2.0.0 has 53 validated source cards. A card records provider, object types, access method, authentication class, rate limits, licence by object, historical depth, earliest public stage, expected-count method, identifiers, deletion behaviour, and source status. The census includes publication-gate platforms, derived review corpora, scholarly identity and outcome services, public funders, and public patent resources. Commercial bibliographic services, paid model APIs, CAPTCHA circumvention, Google Scholar scraping, and endpoints capable of silent paid overages are prohibited dependencies.

The source census covers more providers than the row-level public package. Some sources contribute discovery, reconciliation, pointers, hashes, or validation evidence. Quarantined sources retain the licence, denominator, site-stability, or stage-boundary issue that prevented row release. Each component manifest lists its actual source files.

2.7.2 Publication gates

The publication-gate module covers several distinct layers.

OpenReview. The acquisition system enumerates venue groups, API versions, public submission-state invitations, decisions, withdrawals, desk rejections, reviews, meta-reviews, responses, comments, and revision objects. Invitations are parsed as schemas and permission-bearing workflow objects [3–5]. The final census reconciled five exhaustive shards covering 176 passing venue cycles, 44,330 forums, and 679,898 current Notes. All 176 cycles reconcile their provider-reported and recovered public candidate-state counts exactly. They contain 55,369 observable candidates and 31,992 official reviews; 166 cycles have a non-zero candidate population. The public release distributes non-identifying cycle aggregates and their audit bindings. The raw Note graph remains protected audit evidence. Confidential screening and objects unreadable by the audited free account define these cohorts as Grade B public-stage denominators.

Copernicus and EGU sphere. OAI-PMH, Crossref relations, public discussion pages, and structured article metadata are combined to represent the public discussion stage, referee comments, editor comments, author replies, revisions, and later article relations. The hidden access-review stage remains explicit. Public discussion therefore forms a Grade B cohort conditional on passing that screen.

eLife. Public records are separated into policy-defined editorial-model cohorts. Research Articles and Reviewed Preprints, public reviews, author responses, assessments, evidence/significance terms, version-of-record status, and effective dates are represented where available. “Sent for review” defines the earliest public cohort boundary, after a hidden editorial decision.

F1000-family platforms. F1000Research, Wellcome Open Research, Gates Open Research, and NIHR Open Research are represented through provider-native process records where access and licence checks passed. Editorial screening is separated from post-publication review, versions, report status, responses, and indexing. Evaluation counts exceeded candidate counts in several normalised cycle flows. Those cycles were downgraded under the stage-identity rule.

SciPost. Public submission pages, reports, replies, editorial recommendations, decisions, versions, and arXiv relations are represented. The candidate-pool denominator remains Grade U. SciPost records therefore support process, version, and relation description.

Selected transparent-review histories. PeerJ, PLOS, EMBO Press, Royal Society, and BMC/open-review histories are pointer-first and Grade C by default. Their opt-in or accepted-only records support descriptive review-history research. Qeios and other unresolved sources remain quarantined with their access or denominator failure recorded.

Crossref discovery. Crossref peer-review and work relations provide a broad discovery and relation layer [6]. Depositor and year profiles are retained. Stage completeness comes from separate provider population evidence.

2.7.3 Scholarly identity, text, and outcomes

DOI aliases are canonicalised while preserving one-to-many updates and conflicts. OpenAlex supplies open scholarly identity, citation, and downstream metadata [7]. arXiv contributes version dates, categories, comments, journal references, and DOI links [8]. Europe PMC contributes biomedical identifiers, JATS XML, references, corrections, and retractions where public and licensed [9]. Source-declared identifiers take precedence over bibliographic matching.

The content layer is pointer-first. Redistributable title and abstract metadata and structured content enter release-safe fields. Other objects retain a source pointer, byte or normalised-text hash, extraction recipe, and non-reconstructive features. The public 2.0.0 package excludes full manuscript and review text.

Downstream records distinguish later publication, source-declared repository links, fixed-window citation eligibility, corrections, retractions, withdrawals, and censoring. An unmatched search retains an unknown outcome state.

2.7.4 Research funding as a selection system

The funding component applies the same candidate-cycle-policy model to calls, eligible pools, panel stages, fundable bands, allocations, and awards. This makes the location of a public population boundary directly comparable with publication review while preserving the distinction between scientific assessment and budget-constrained allocation. It combines public UKRI opportunity and meeting-round records with frozen FWF and SNSF award data [15,16] and an instrument-level evaluability census. The release represents 18 instruments (`funding_instruments`), 2,102 UKRI opportunities, 1,449 public UKRI panel-round aggregates, and 1,555 repeated-awardee lower-bound records. These are different units.

The panel-population extension adds candidate-level choice sets to the award records. Official EPSRC and AHRC outcome workbooks yield 21,883 application-panel events in 1,521 meeting/list choice sets containing both funded and unfunded or otherwise resolved events. Applications appearing at more than one panel remain separate events linked to the same protected application identifier. Exact duplicate source rows are collapsed with their count recorded. Names and proposal text are excluded from the release.

A second open component reconstructs nine anonymised SNSF panels from the individual-vote workbook deposited with the published evaluation study [23]. It contains 432 proposal-panel events and 4,811 expected proposal-voter cells. Recorded grades, explicit conflicts of interest, and blank cells are separate states. A subset of panels reports funding outcomes for 353 proposals. Proposal labels are panel-specific, and blank cells retain their source state.

Each instrument is evaluated against six disclosures: application counts, the passing or eligible pool, assignment arm, awards or outcomes, unfunded candidate text, and decision date or round. Registered estimands are then allowed or denied deterministically. Awardee recurrence is labelled as a lower bound on repeated awardees. Application recurrence requires additional submission histories. The UKRI and SNSF tables identify observed panel-stage choice sets. Their scope supports panel composition, ranking, voting, and descriptive allocation research; pre-panel entrants and randomised assignment arms remain outside the public data.

2.7.5 Patent examination as a selection system

Patent prosecution supplies another contrasting process architecture. An application can receive repeated office actions, undergo claim amendment, and move among legal grounds before allowance or abandonment. OSG maps those turns to the same candidate-event graph used for other gates. Legal novelty and

scientific novelty retain separate definitions. The process component uses the public PANORAMA release and USPTO research-data documentation [17,18] to represent 8,143 applications with an observable non-final rejection and allowance trail. The derived products contain 40,715 stage-chain rows, 174,120 claim-alignment rows, 24,232 examiner-cited patent links, and 3,391 art-unit-year capacity cells. Claim and office-action text are excluded from the release.

Legal grounds are separated: novelty under 35 U.S.C. §102, non-obviousness under §103, eligibility under §101, disclosure or clarity under §112, and other actions. Claim comparison uses explicit state accounting. Same-number claims cover 0.909 of the comparison estimand; 229 initial-only and 15,548 final-only claims are explicit transition states, giving complete accounting over all 174,120 rows. Cross-number alignment requires separate evidence.

To locate this process sample within a much larger public population, OSG reduces the revision-pinned HUPD metadata release [24] to 4,518,254 distinct public English-language US utility applications filed from 2004 through 2018. The release retains the full 4,518,254-row privacy-minimised application census, 578 aggregate population cells, and a hash-only PANORAMA crosswalk; it excludes invention text, examiner names, docket and confirmation numbers, and the unredacted upstream HUPD rows. The join recovers 91.49% of PANORAMA cases filed from 2004 through 2016. Later PANORAMA filings carry an outside-HUPD-time-boundary state.

Every PANORAMA case has exactly one reconciliation state: 3,416 exact application-number matches, 1,195 cases inside HUPD’s filing-year boundary and absent from that frozen population, and 3,532 cases outside the boundary. The three states account for every PANORAMA row. HUPD and PANORAMA have different temporal and coverage boundaries.

Applicant response documents remain absent. Claim-set changes after an office action provide a timing proxy. Response text, authorship, and argument content are outside the released data, as are confidential and non-published applications.

2.7.6 Scope boundaries

The release distinguishes broad source coverage from populations that support statistical rates. The table below summarises the realised evidence and the corresponding admissible interpretation.

Table 2. Realised scope and admissible interpretation.

Resource layer	Realised state in version 2.0.0	Admissible interpretation
Public-source census	53 source cards; included, pointer-only, and quarantined states retained	Exhaustive documented census of attempted public sources; row inclusion varies by component
OpenReview process census	Five-shard census passed for 176 public cycles; raw Note graph retained as audit evidence, filtered derived tables released	Stage-conditional analysis only where cycle-specific coverage passes
Observability census	4,872 rows, including 4,522 policy/configuration surfaces	Institutional-opacity and source-coverage research; policy-only rows are excluded from rates
Process-level analytical atlas	350 reconstructed cycles; 210 verified A/B denominators; 200 populated	61 complete observed selection-rate cycles and 197 review-incidence cycles
Many transparent-review providers	Several providers remain selected-only, pointer-only, or unresolved	Review-process description, with candidate-pool selection restricted to verified cohorts
Multi-ruler novelty	Pinned SPECTER2 and Qwen3 embeddings over 8,501 shared time-valid versions; citation recombination for 93 versions and textual recombination for 7,939	Plural measurement with ruler disagreement and ruler-specific coverage
Cross-gate lineage	1,080 source-declared edges, 8,627 chain rows, and a separately scored probabilistic candidate tier	High-precision trajectory analysis or user-selected recall sensitivity; canonical merges use the source-declared tier

Table 2. Realised scope and admissible interpretation (continued).

Resource layer	Realised state in version 2.0.0	Admissible interpretation
Public funding choice sets	21,883 UKRI application–panel events and 432 SNSF proposal–panel events with individual votes	Panel-stage population and voting research within the observed choice sets
Patent examination	8,143-case PANORAMA process panel benchmarked against aggregate cells derived from 4,518,254 published English-language US utility applications filed from 2004 through 2018	Process analysis within declared public, filing-year, language, and application-type boundaries

Table 3. Process-level scope and denominator-dependent analytical eligibility. Policy-only configuration rows are retained in a separate observability census.

Platform	Process cycles	Grade A/B	Populated A/B	Selection-rate	Review-rate
OpenReview	181	176	166	27	163
F1000 family	36	26	26	26	26
Royal Society	26	0	0	0	0
Copernicus	24	0	0	0	0
BMC	18	0	0	0	0
PLOS	16	0	0	0	0
SciPost	15	0	0	0	0
PeerJ	13	0	0	0	0
eLife	8	8	8	8	8
EMBO	7	0	0	0	0
Qeios	5	0	0	0	0
Crossref	1	0	0	0	0
Total	350	210	200	61	197

Table 4. Verified OpenReview public-process cohort. Review counts are complete for the audited public Note graph; they do not include confidential or unreadable objects.

Quantity	Value
Audited cycles	176
Populated cycles	166
Observable candidates	55,369
Official reviews	31,992
Cycles with at least one official review	38
Mean reviews per cycle	181.8
Median reviews per cycle	0.0

2.8 Data model and identifiers

2.8.1 Core graph

The canonical model contains 24 schema artefacts plus a SQL schema and catalogue. Its central relationships are summarised below; each arrow denotes a one-to-many relation unless the schema records a many-to-many link explicitly.

```

gate → gate_cycle → policy_version
gate_cycle → coverage_observation, capacity_observation
gate_cycle ↔ candidate_gate_event ↔ candidate
candidate → candidate_version → evaluation, decision_event, content_artifact
candidate → lineage_edge, downstream_outcome
source_object → provenance_event

```

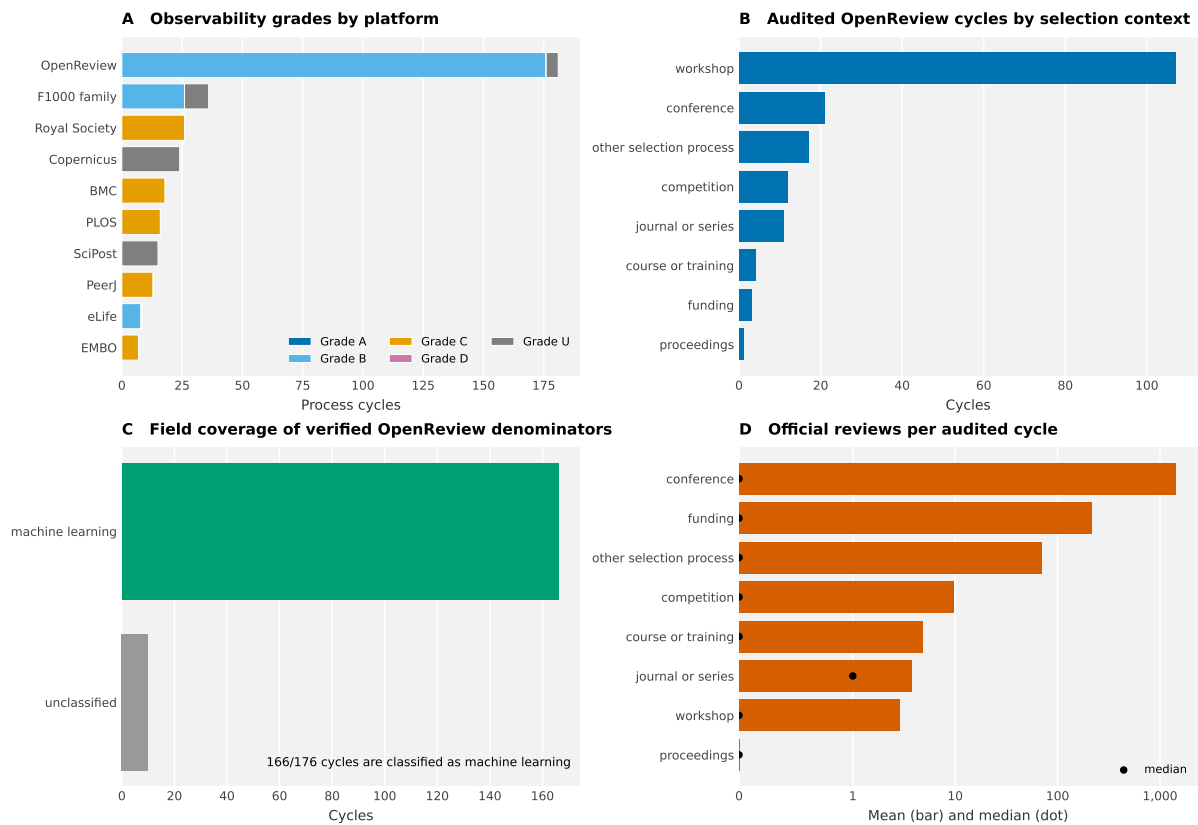


Figure 1. Scale and composition of the process-level analytical release. Panel A separates observability grades by platform. Panels B–D describe the 176-cycle provider-audited OpenReview cohort by selection context, field coverage, and official-review volume. Review volume is concentrated in three large ICLR conference cycles. The figure therefore reports means and medians; the median across all audited cycles is zero, and 38 cycles contain at least one official review.

Additional tables represent identifier aliases, authorship observation, identity visibility, fields, references, language derivatives, and field-level provenance. Canonical IDs are deterministic namespaced UUIDv5 values. Stable native identifiers are inputs where available; content hashes or deterministic compound keys are used otherwise. Reharvesting preserves the ID. Uncertain same-work relations remain evidence-bearing edges with direction, confidence, and linkage tier.

2.8.2 Provenance and storage

Raw objects are immutable, compressed, and content-addressed. Retrieval manifests record source, native identifier, query, retrieval time, byte hash, connector version, and checkpoint state. Large inactive raw sources can be packed into append-only indexed containers only after every member is verified before and after packing. Normalised data are partitioned Parquet; DuckDB exposes append-only history views and deterministic current views.

Every normalised fact links to a `source_object_id` and `provenance_event_id`. Coherent field groups additionally link through `field_provenance` to a source path or JSON pointer, parser version, confidence, and override history. Tombstones record the prior existence of a removed object and omit its payload.

2.8.3 Acquisition and checkpoints

All adapters implement the same contract: discover, count, fetch, checkpoint, normalise, validate a fixture, and emit coverage. Supported transports include REST cursors and pages, OAI-PMH, JSONL/CSV bulk data, XML/JATS, HTML, sitemaps, and PDFs. Per-host concurrency, cache reuse, conditional requests, `Retry-After`, bounded exponential backoff, daily request ceilings, and explicit user-agent identification are mandatory.

Before a population pull, an adapter must pass a deterministic fixture, source-card, terms/licence review, independent count method, dry run, and resource estimate. Checkpoints atomically retain the source query, cursor or page token, last native ID, row count, and shard hash. Forced interruption tests compare

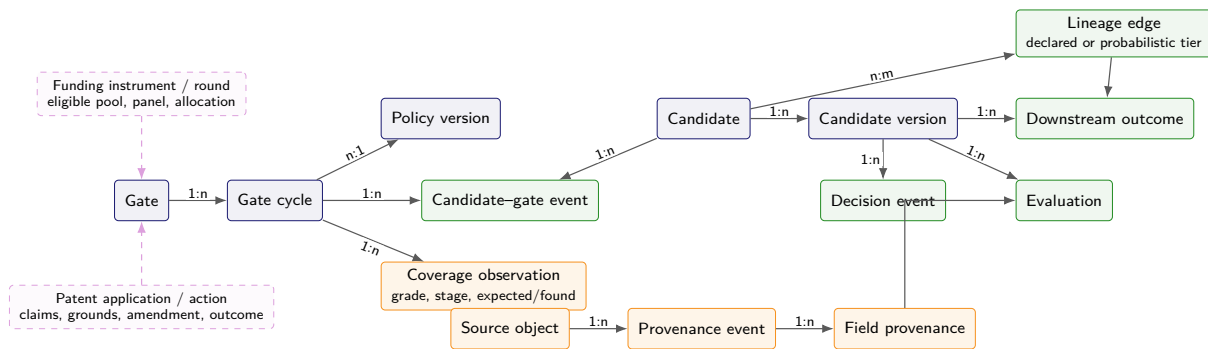


Figure 2. Entity-relationship view of OSG. Candidate-gate events connect research objects to institutional cycles; coverage observations determine which denominators and rates are admissible. Funding rounds and patent prosecution are represented as domain-specific realisations of the same gate-candidate-event pattern without erasing their native stages or legal meanings. Every normalised fact remains traceable to source and field-level provenance.

restarted output with uninterrupted canonical output.

OpenReview required additional safeguards because venue semantics are invitation-specific and the public APIs expose both current Notes and Edit histories. The final acquisition used five disjoint domain-history shards and five contiguous current-Note forum ranges. A union hash, per-shard run manifests, independent provider totals, and a forum sample were verified before inclusion in the analytical atlas.

2.9 Derived products

2.9.1 Institutional regimes and flow series

The institutional archive contains one policy observation for each of 4,696 locally reconstructed gate-cycle rows. An explicit unknown observation represents the absence of a dated policy. Current undated pages support current-state description only. The historical archive contains 4,531 policy-version rows. Machine-readable OpenReview configuration contributed 4,525 rule records, and the rubric crosswalk contains 11,188 native criterion-field rows.

The flow series adds 176 provider-audited OpenReview cycles and records observable, reviewed, withdrawn, rejected, selected, and unresolved counts together with provider expected/found counts, coverage grade, hidden-stage note, and identity checks. Candidate-denominator validity, review-incidence eligibility, and selection-rate eligibility are separate fields. Stage-flow violations cause exclusion or downgrade.

2.9.2 Evaluations and construct atlas

The release contains 40,483 normalised evaluation objects, of which 40,102 join to an exact candidate version and gate. Evaluation kinds include peer reviews, editorial or meta-evaluations, and other evaluation objects. Official, non-official, and unspecified statuses are separate fields. Public comments retain their native classification.

Native rubric values round-trip exactly. Within-cycle calibrated values are additional fields, and native scales remain available. The construct layer represents 5 broad human-evaluation constructs and 4 available ruler families in 9 selector rows. Construct-bearing spans are released for explicit native criterion labels. A deterministic research-team audit assessed 314 such labels with observed precision 1.0 and a Wilson 95% lower bound of 0.988.

Free-form interpretation is evaluated separately against Peer Review Analyze 1.0 [22]. A frozen Qwen3 representation and class-balanced logistic models are evaluated in five folds grouped by submission over 13,724 human-annotated spans. The five construct AUCs range from 0.72 to 0.938. Only out-of-fold probabilities and metrics are retained. Review prose and embeddings are excluded from the release and from training of the explicit-rubric crosswalk. The AUC range describes external benchmark performance on ICLR 2018 annotations. OSG review records carry no free-form construct score.

2.9.3 Semantic and recombinatorial novelty

Semantic novelty is represented by two independently trained embedding rulers. AllenAI’s SPECTER2 scientific-document encoder [20] is pinned to commit 3447645e1def9117997203454fa4495937bfbdb83. The general-purpose Qwen3-Embedding-0.6B model [21] is pinned to commit 97b0c614be4d77ee51c0ce

f4e5f07c00f9eb65b3. Both encode the same title and abstract input. Fitting and model selection are independent of OSG outcomes. Each target year is compared with documents created before 1 January of that year. Each table reports cosine distance to the prior centroid and nearest prior document, normalised within source-year cells. Manifests bind the model revision, input and reference-set hashes, embedding dimension, parameters, and output checksum. Dense vectors and input text are removed after feature verification.

SPECTER2 yields 8,516 rows and Qwen3 yields 8,507; 8,501 versions have both rulers. Their within-source-year centroid percentiles correlate at 0.554. The median absolute percentile disagreement is 0.167. The release supplies both values and their disagreement as separate fields. SPECTER2 contributes a citation-informed scientific training distribution; Qwen3 contributes an independently trained, instruction-aware text representation. Their agreement quantifies convergence between rulers. Another 450 future-dated records remain in metadata and are excluded from time-prior corpora.

Recombinatorial novelty follows the prior-combination logic used in science-of-science research [19], operating on resolved cited-work or identifier pairs. For each target, pair frequencies are computed from prior years only; the release retains median conventionality, lower-tail novelty, and new-combination share. 93 candidate versions have adequate citation-pair data.

A second recombination ruler operates on deterministic document-internal lexical elements: normalised tokens and adjacent token pairs. Its vocabulary-free construction uses no outcome labels. For each year, element-pair frequencies are drawn only from earlier documents. It measures 7,939 of 8,563 eligible versions (92.71%). The public table contains counts and scores without phrase text. Citation and textual recombination remain separate columns because they describe different opportunity spaces. Their overlap measures convergence.

Journal-pair and ontology-concept variants require validated journal and concept resolution, which is absent from the current public reference records. Their fields retain missing values.

2.9.4 Lineage and versions

Lineage edges preserve native relation, normalised relation class, direction, declaration status, confidence, evidence, and linkage tier. Relations include new version, resubmission, transfer, split, merge, extension, and related-only. Many-to-many structures are allowed.

The high-precision analysis layer contains only validated source-declared relations. It achieved observed precision 1.0 on 711 audited edges with a Wilson 95% lower bound of 0.995. Candidate generation recovered all source-declared benchmark pairs in the frozen benchmark. Title collisions remain candidate edges. A parallel recall-oriented tier begins with each version's 11 nearest titles under the revision-pinned Qwen3 encoder and scores title, abstract, source, and time features. Source-declared relations form positive examples; similar non-declared pairs form explicitly labelled proxy hard negatives. Among 75,897 scored version-pair candidates, 6,652 pass the declared-positive recall threshold, 3,330 pass the balanced threshold, and 253 pass the high-score threshold. These proxy-labelled scores serve discovery and sensitivity analysis. Validated source-declared relations define the analysis-grade lineage tier.

Human evidence is supplied in a separate calibration table from the PreprintToPaper Gray Zone [25]. Two annotators gave decisive labels to 284 ambiguous pairs and agreed on 269 ($\kappa = 0.859$). A five-fold out-of-fold model at the study's title-similarity design point achieves an AUC of 0.971 using author overlap. The current OSG public tier lacks that author feature, so the calibration probabilities remain benchmark-only. This result also identifies author overlap as a requirement for future transfer.

Self-loops, temporally impossible edges, and graph cycles are quarantined. The release also contains 1,080 source-declared lineage rows, 199 aligned version pairs, and 8,627 candidate-gate chain rows.

2.9.5 Downstream observation states and fixed windows

The downstream panel contains 8,903 candidate rows. It separates source-declared later-publication edges, public repository links, corrections/withdrawals, fixed-window citation observations, and censoring. Publication-link validation achieved observed precision 1.0 with a Wilson 95% lower bound of 0.989. Citation windows are emitted for mature rows. 113 immature rows retain an ineligible state. The observation-window table expands these candidates into 35,612 rows at one-, two-, three-, and five-year horizons, of which 32,510 meet the corresponding follow-up requirement. These rows describe what was observable by the snapshot. Exact publication dates are unidentified, so the table supports fixed-window analysis and excludes survival-time estimands. Publication linkage uncertainty is represented by 224 source-window bounds and capture-fraction scenarios. All 8,903 candidates without an exact publication

event time retain an unknown publication state.

2.9.6 Strain, reforms, and architecture comparisons

Timing products retain UTC-normalised submission, review, response, decision, revision, and delay measures where timestamps exist; missing timestamps remain rows with reasons. Evaluator-supply and workload proxies are aggregated to gate-cycle level. The public release contains aggregate evaluator measures only. Three policy events satisfy the frozen reform registry. Other policy differences remain descriptive.

The construct model compares a single common factor, a correlated multi-trait representation, and a method-separated representation over complete cases. Reported correlations are uncorrected for reliability, and stages remain separate. Existing rows are exploratory. Future confirmatory tests use the frozen holdout registry.

3 Data records and release organisation

Version 2.0.0 is an immutable release snapshot. It contains 72 Parquet tables in 6 licence-separated components. Tables have different units, and several are views or derived measurements over the same underlying objects. Their row counts therefore describe each table individually.

Table 5. Release components, licences, contents, and exclusions.

Component	Licence	Contents	Principal exclusions
metadata	CC0-1.0	coverage atlas, gate-cycle flows, policy archives, rubrics, schemas, documentation, benchmark definitions, static explorer	raw source objects and protected identifiers
derived_features	CC BY 4.0	evaluations, semantic/recombinatorial novelty, lineage, downstream observation states, version alignment, strain, construct, partial-identification, and transport products	source full text, review text, reconstructive restricted-text features
funding_open_panels	CC BY 4.0 / OGL 3.0	UKRI application-panel events and SNSF proposals, votes, and panel outcomes	names, proposal text, pre-panel entrants, assignment arms
funding_noncommercial	CC BY-NC-SA 4.0	instrument evaluability, public opportunity/panel/portfolio series, repeat-awardee lower bounds, grant-output links	applicant microdata, unfunded proposal text, inferred assignment arms
patent_noncommercial	CC BY-NC-SA 4.0	PANORAMA application, action-chain, claim-alignment, prior-art and capacity tables plus HUPD population cells and a hash crosswalk	claim/action text, confidential applications, personnel rankings
validation	CC0-1.0	machine-readable build, coverage, linkage, leakage, privacy, licence, and recovery reports	secrets and protected audit identities

Each package contains `VERSION.json`, `VALIDATION.json`, `PACKAGE.json`, a file-level SHA-256 manifest, schemas, table counts, and a DuckDB database. Reuse follows the licence assigned to each component. The deposit bundles are deterministic tar archives with normalised ownership and timestamps. The DuckDB database materialises the smaller analytical tables. Very large censuses, including the full released HUPD boundary population, remain directly queryable Parquet assets and are enumerated with row counts and paths in `PACKAGE.json`; the release verifier checks both storage modes.

4 Technical validation

4.1 Schema, invariants, and coverage

Primary and foreign keys, enum domains, temporal ordering, policy intervals, native/normalised pairing, provenance reachability, licence flags, stage flows, and graph cycles are validated on fixtures and release tables. Coverage requires an independent expected-count method for any A/B population. The source

atlas freezes after every registered acquisition source has coverage evidence and every required export is present.

The stage/outcome audit checked 7,255 supported mappings across stratified native outcomes and found observed precision 1.0, above the prespecified release threshold. Unsupported values remain native-only. Policy extraction is audited against dated raw source evidence; configuration hash, architecture, criteria, effective date, byte hash, and scale must all agree. Historical change claims use dated policy evidence.

4.2 Temporal leakage

The temporal validator tests candidate features against decision time, reference cutoff, identity visibility, and source observation time. Deliberately contaminated fixtures include final versions, future citations, and post-decision identity. They must fail. The semantic builder separately excludes target-year records from their reference corpora and rejects future-dated source metadata from feature eligibility.

4.3 Linkage and missingness

Source-declared links are the highest-precision benchmark. Probabilistic candidates are released separately with three documented thresholds. Canonical merges use source-declared evidence. Edge and trajectory counts can be recomputed under strict source-declared, probabilistic high-recall, balanced, and high-score layers. Missing text, references, evaluations, outcomes, and identities are profiled by source and format. Metadata rows remain when content is absent, and every feature family declares its coverage rule.

Two external human benchmarks test the places where source declarations alone are insufficient. Peer Review Analyze supplies construct labels for free-form prose; all predictions are out of fold and grouped by submission. PreprintToPaper supplies independently annotated ambiguous lineage pairs, inter-annotator agreement, and a calibrated author-overlap model. OSG stores these benchmark predictions in separate tables from the source-declared tiers. Source-declared evidence governs canonical entities.

Funding and patent extensions have independent population checks. The UKRI builders reconcile duplicate rows, outcomes, repeated panels, and choice-set totals against the official workbooks. The SNSF strict-OOXML parser preserves every expected proposal-voter cell and validates the six-value grade domain. The patent builder reconciles all 8,143 PANORAMA cases, performs complete claim-state accounting, and audits HUPD overlap within the shared filing-year window before publishing the crosswalk.

4.4 Determinism and clean-room recovery

Semantic novelty, recombinatorial novelty, lineage, version alignment, and fixed-window outcome artefacts were rebuilt and compared by exact SHA-256 in the determinism audit. Clean-room recovery reconstructs registries, schemas, and representative public fixtures from the released repository in an isolated environment. The test uses no network access, credentials, pre-existing generated outputs, or protected raw lake. Full-source recovery is tested separately through content hashes, retrieval manifests, checkpoints, and provider-specific refetch recipes. A disaster-recovery drill verified 26 schema artefacts, 53 source cards, 21 registered estimands, and a representative seven-source normalised partition.

4.5 Privacy, licensing, and security

The release scanner checks table schemas and packaged paths for contact columns, evaluator identifiers, protected authorship identifiers, raw/cache paths, and restricted content. An adversarial privacy and licensing audit tested contact leakage, reviewer re-identification joins, restricted-text reconstruction, licence incompatibility, and deleted-object recovery. All highest-severity findings were resolved before release. Removal simulations propagate a tombstone through normalised objects, features, views, manifests, and changelogs.

Acquisition credentials enter through ephemeral environment or provider secret stores. The framework redacts authorisation data and scans repository history and artefacts. Public-output reproduction uses no credentials.

5 Usage notes

5.1 Choosing a denominator

Users should begin with `gate_cycle_descriptive_atlas` or `coverage_observation`. An entry-selection rate requires Grade A and all entrants at the gate-entry stage. A stage-conditional rate may use Grade

A or B and must name the public stage. Grade C supports statements about visible or selected review histories. Grade D supports portfolio description. Grade U supports source auditing and missingness research.

Conceptually, a valid stage-conditional query is:

```
SELECT gate_cycle_id,
       selected_count::DOUBLE / observable_count AS selected_share
FROM gate_cycle_descriptive_atlas
WHERE effective_observability_grade IN ('A', 'B')
      AND descriptive_rate_allowed
      AND observable_count > 0;
```

The result must be described as conditional on the row's `earliest_public_stage`. The verified observable population supplies the denominator.

5.2 Evaluation and novelty research

Evaluation analyses should join `evaluation_objects` to the exact candidate version and inspect `official_status`, native rubric, scale, and policy version. Construct comparisons should join their reliability ceilings and report uncorrected estimates. Semantic features should name encoder, model revision, reference cutoff, and source-year normalisation. When conclusions depend on semantic distance, users should report SPECTER2, Qwen3, and their disagreement. Citation and textual recombination require separate coverage statements. Every novelty measure should name its construct and reference corpus.

5.3 Lineage and afterlife research

Within-work analyses should use the source-declared validated layer for high-precision estimates. Studies prioritising recall may select a published probabilistic threshold, report sensitivity across thresholds, and propagate linkage uncertainty. Later acceptance reflects endogenous routing, withdrawal, revision, and survival. Final-version features are post-treatment relative to an earlier rejection. Unmatched candidates retain their right-censored state.

5.4 Funding and patent research

Funding estimators should call the evaluability firewall before use. The UKRI and SNSF proposal tables support panel-stage comparisons. Their pre-panel populations and randomised allocation arms are unobserved. Winner-only records support portfolio and award-recurrence description. Patent process analyses are conditional on PANORAMA's rejection-to-allowance sample. Population comparisons must additionally impose the HUPD filing-year, language, publicity, and utility-application boundary. Legal grounds must remain separate. Capacity is released at art-unit-year level with small-cell suppression; personnel ranking is prohibited.

5.5 Community benchmarks

Six non-generative benchmark tasks cover observability classification, policy/rubric extraction, stage normalisation, version linkage, construct measurement, and calibrated missingness. They use source, time, or domain holdouts; exclude final-outcome leakage and reviewer identities; and require licence-cleared fixtures or rebuild pointers. Scoring depends only on released or freely rebuildable inputs.

6 Ethical and legal considerations

The project uses public data and involved no recruitment, contact, intervention, or bespoke agreement with human subjects. Public availability establishes an acquisition condition. Redistribution and permitted analyses follow object-level licences and privacy rules.

Anonymous or pseudonymous reviewers face the highest re-identification risk. Named authors, applicants, editors, and decision makers face ranking, employment, harassment, and targeting risks. Public packages therefore exclude evaluator identifiers, direct contact fields, protected authorship identifiers, raw source objects, and purpose-irrelevant person-level joins. Evaluator-supply measures are aggregate. The project prohibits reviewer deanonymisation, identity-inference joins intended to reveal anonymous persons, individual productivity or quality ranking, employment or funding evaluation, harassment, surveillance, automated targeting, and restricted-text reconstruction.

Licensing is object-specific. Metadata, article text, review text, responses, policy documents, images, and derived features may carry different rights even within one provider. The release builder selects redistribution, pointer/hash, derived-only, or exclusion per object and separates incompatible components. A hash records provenance. Rights to retrieve or reconstruct content follow the source licence.

Corrections and takedowns create versioned tombstones and amended releases. Source platforms and third-party archives retain their own records. Community corrections enter through evidence-bearing issue templates, and every change to a canonical fact creates a new provenance event.

7 Limitations

OSG describes populations that become visible in public records. Hidden screens create selection into observation. No-assumption regions and capture-fraction scenarios quantify the compatible entry populations; narrower estimates require a stated substantive assumption.

Source populations vary in architecture, calendar period, size, selection share, and review intensity. The transport table measures overlap in these dimensions for every source pair. Domain coverage is incomplete, and cycle-level policy and disclosure covariates are absent. The current diagnostics therefore support conditional modelling or descriptive stratification according to the recorded verdict.

Licence and privacy rules separate the public package from the protected acquisition lake. The pointer registry makes 684 omitted objects verifiable under their source terms. Retrieval and redistribution remain governed by the provider and object licence.

Semantic and recombinatorial measurements depend on their construct and reference corpus. SPECTER2 and Qwen3 show moderate rank agreement. Textual and citation recombination describe different opportunity spaces. OSG releases the separate scores, their coverage, and their disagreement. Free-form construct probabilities are confined to the external ICLR 2018 benchmark; transfer to other fields, languages, or review architectures requires new validation.

Source-declared lineage defines the analysis-grade tier. The PreprintToPaper benchmark includes author overlap, a feature absent from the current OSG public tier. Qwen3 candidate scores consequently remain a discovery and sensitivity product.

Publication observation windows record maturity, right censoring, competing withdrawal states, and linkage uncertainty. Exact publication dates are unavailable, so these tables support fixed-window analyses. Publications absent from every linked database remain unknown outcomes.

The funding extension covers funded and unfunded panel-stage candidates and individual votes for the included UKRI and SNSF panels. Pre-panel applicants, proposal text, and randomised assignment are outside its scope. The repeated-awardee table gives a lower bound on award recurrence.

The privacy-minimised HUPD census supplies a comparison population of 4,518,254 published English-language US utility applications filed from 2004 through 2018. The crosswalk records exact matches and boundary differences between HUPD and the PANORAMA process panel. PANORAMA accounts for claim-set transitions. Claim-set change supplies a timing proxy because applicant response documents are absent. Legal novelty and scientific novelty remain distinct constructs.

8 Code availability

The release is generated from public or genuinely free sources without paid data or model APIs. Adapter, normalisation, validation, release, and benchmark code is included in the code/metadata component. The canonical validation and rebuild commands are:

```
PYTHONPATH=src .venv/bin/python -m observatory.cli validate
PYTHONPATH=src .venv/bin/python -m observatory.cli r1-build
PYTHONPATH=src .venv/bin/python -m observatory.cli r2-build
PYTHONPATH=src .venv/bin/python -m observatory.cli r3-build
PYTHONPATH=src .venv/bin/python -m observatory.cli r4-build
PYTHONPATH=src .venv/bin/python -m observatory.cli r5-build
PYTHONPATH=src .venv/bin/python -m observatory.cli validity-build
PYTHONPATH=src .venv/bin/python -m observatory.cli release-validate \
  --release-id open-selection-graph-2.0.0
PYTHONPATH=src .venv/bin/python -m observatory.cli release-build \
```



```
results/observatory/releases/open-selection-graph-2.0.0
PYTHONPATH=src .venv/bin/python -m observatory.cli publication-bundles
```

The immutable package is identified by release version 2.0.0, source snapshot 2026-08-23, the version-registry snapshot identifier, the package manifest, and file-level SHA-256 hashes. These identifiers allow any copy of the package to be verified independently.

9 Data availability

The data records described in this article are contained in the licence-separated release package archived at <https://doi.org/10.5281/zenodo.22081233>. The code and metadata are also available at <https://github.com/anton-lw/open-selection-graph>, and the release archives are mirrored at <https://huggingface.co/datasets/antonsaccount/open-selection-graph>. The package-level `VERSION.json`, `PACKAGE.json`, `VALIDATION.json`, and `MANIFEST.json` files identify the snapshot and provide table counts, component licences, and SHA-256 checksums. Pointer-only source material must be retrieved from its original provider under the provider's terms; the release records the source identifier, retrieval recipe, and content hash needed to verify such reconstructions.

10 Acknowledgements

OSG depends on public infrastructure and upstream datasets maintained by OpenReview, Crossref, OurResearch/OpenAlex, arXiv, Europe PMC, Copernicus Publications, eLife, F1000-family platforms, SciPost, transparent-review publishers, public research funders, the United States Patent and Trademark Office, and the creators of the derived corpora credited in the release attribution manifest. Upstream authorship and licences are preserved in machine-readable and human-readable attribution files.

11 References

- Kang, D. et al. A Dataset of Peer Reviews (PeerRead): Collection, Insights and NLP Applications. *Proceedings of NAACL-HLT 2018*, 1647–1661 (2018). <https://doi.org/10.18653/v1/N18-1149>
- Dyke, N., Kuznetsov, I. & Gurevych, I. NLPeer: A Unified Resource for the Computational Study of Peer Review. *Proceedings of ACL 2023*, 5049–5073 (2023). <https://doi.org/10.18653/v1/2023.acl-long.277>
- OpenReview. Introduction to Notes. <https://docs.openreview.net/getting-started/using-the-api/objects-in-openreview/introduction-to-notes> (accessed 20 August 2026).
- OpenReview. Introductions to Invitations. <https://docs.openreview.net/getting-started/using-the-api/objects-in-openreview/introductions-to-invitations> (accessed 20 August 2026).
- OpenReview. Introduction to Groups. <https://docs.openreview.net/getting-started/objects-in-openreview/groups> (accessed 20 August 2026).
- Crossref. REST API. <https://api.crossref.org> (accessed 20 August 2026).
- Priem, J., Piwowar, H. & Orr, R. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv* 2205.01833 (2022). <https://doi.org/10.48550/arXiv.2205.01833>
- arXiv. Bulk Data Access. https://info.arxiv.org/help/bulk_data.html (accessed 20 August 2026).
- Europe PMC. Developers. <https://europepmc.org/developers> (accessed 20 August 2026).
- Copernicus Publications. Public peer review and interactive public discussion. https://publications.copernicus.org/services/public_peer_review.html (accessed 20 August 2026).
- eLife. Peer review. <https://elifesciences.org/about/peer-review> (accessed 20 August 2026).
- F1000Research. How it works and developer resources. <https://f1000research.com/about> and <https://f1000research.com/developers> (accessed 20 August 2026).
- SciPost. Editorial procedure and public submissions. <https://scipost.org/submissions/> (accessed 20 August 2026).
- PLOS. Published Peer Review History. <https://plos.org/resource/open-peer-review/> (accessed 20 August 2026).
- Austrian Science Fund. FWF Open API. <https://openapi.fwf.ac.at> (accessed 20 August 2026).
- Swiss National Science Foundation. SNSF Data Portal. <https://data.snf.ch> (accessed 20 August 2026).
- LG AI Research. PANORAMA patent office-action dataset. <https://huggingface.co/datasets/LG-AI-Research/PANORAMA> (accessed 20 August 2026).
- United States Patent and Trademark Office. Patent examination and office-action research datasets. <https://www.uspto.gov/ip-policy/economic-research/research-datasets> (accessed 20 August 2026).
- Uzzi, B., Mukherjee, S., Stringer, M. & Jones, B. Atypical combinations and scientific impact. *Science* **342**, 468–472 (2013). <https://doi.org/10.1126/science.1240474>
- Singh, A., D'Arcy, M., Cohan, A., Downey, D. & Feldman, S. SciRepEval: A Multi-Format Benchmark for Scientific Document Representations. *Proceedings of EMNLP 2023*, 5548–5566 (2023). <https://doi.org/10.18653/v1/2023.emnlp-main.338>
- Zhang, Y. et al. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. *arXiv* 2506.05176 (2025). <https://doi.org/10.48550/arXiv.2506.05176>
- Ghosal, T., Kumar, S., Bharti, P. K. & Ekbal, A. Peer review analyze: A novel benchmark resource for computational analysis of peer reviews. *PLOS ONE* **17**, e0259238 (2022). <https://doi.org/10.1371/journal.pone.0259238>
- Heyard, R., Ott, M., Salanti, G. & Egger, M. Rethinking the Funding Line at the Swiss National Science Foundation: Bayesian Ranking and Lottery. *Statistics and Public Policy* (2022), with individual vote data at <https://doi.org/10.5281/zenodo.4531160>. <https://doi.org/10.1080/2330443X.2022.2086190>
- Suzgun, M., Melas-Kyriazi, L., Sarkar, S. K., Kominers, S. D. & Shieber, S. M. The Harvard USPTO Patent Dataset: A Large-Scale, Well-Structured, and Multi-Purpose Corpus of Patent Applications. *arXiv* 2207.04043 (2022). <https://doi.org/10.48550/arXiv.2207.04043>
- Badalova, F., Sienkiewicz, J. & Mayr, P. PreprintToPaper dataset: connecting bioRxiv preprints with journal publications. *Scientific Data* (2026). <https://doi.org/10.1038/s41597-026-06867-3>